

Hybrid Architecture PRD: Trait-Driven Scientific Computing System

Version 12 — Static Specialization as Primary, Graph IR for Dynamic Concerns

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§ Abstract

This document specifies a multi-paradigm scientific computing system built in Mojo. The system unifies algebraic number types (Clifford, exterior, dual, jet, quaternion, and others), automatic differentiation, numerical integrators, and backend execution under a single coherent architecture. The core architectural thesis is:

Traits + AlgebraSpec + Comptime Specialization + Graph for Dynamic Concerns

This is a trait-driven, statically-specialized system. The primary execution path generates specialized kernels at compile time from method specifications and algebra specifications. A

runtime graph IR exists only for dynamic orchestration and reverse-mode AD.

CONTENTS

- 01 Vision & scope
- 02 Mathematical foundations
- 03 Trait layer specification (the center of gravity)
- 04 Algebra specs & generated types
- 05 Compilation model: static / planned / dynamic three-tier execution
- 06 Compile-time IR & optimization passes
- 07 Runtime graph: scope & boundaries (explicitly narrowed)
- 08 AD architecture: forward algebraic + reverse graph
- 09 Backend & storage
- 10 Validation stack (property tests, Lean export, Mathlib4 alignment)
- 11 Implementation roadmap (16 milestones)
- 12 Appendix A — Code examples and benchmarks
- 13 Appendix B — Mojo alignment notes
- 14 (plus seven more appendices)

§ The architectural resolution

The PRD resolves a long-running debate between a graph-IR-first approach and a comptime-first approach in favor of a three-tier model (static / planned / dynamic). Key structural decisions:

- **GLMSpec with coefficient matrices (A, U, B, V) as the unifying integrator abstraction** — subsumes RK, BDF, multistep, and peer methods.
- **A domain-specific CAS built from the same AlgebraSpec building blocks** feeding a CAS → Lean → comptime verified pipeline.

- **Algebraic lifting as a general enrichment mechanism** covering AD, covariance propagation, polynomial chaos, and interval validation.
- **Comptime-generated orthogonal polynomial families** (Hermite, Legendre, Laguerre, Chebyshev, Jacobi) enabling PCESpec as a legitimate AlgebraSpec for uncertainty quantification.

§ Scoping decisions

- Orbit-type-aware propagation and polynomial event detection belong in *consumer packages*, not the core library.
- The CAS should be *narrow and focused* — exact rational arithmetic, spec-derived rewriting, GLM coefficient derivation, Lean export — not a general-purpose CAS.
- PDE support is a consumer layer built on top of the core.
- The library is positioned as a *backbone for astrodynamics tools* (STK / GMAT / Orekit / heyoka successors) and high-fidelity scientific simulation, not for real-time physics engines like NVIDIA Newton.

§ Reference implementations studied

DifferentialEquations.jl (problem/algorithm separation, sensitivity-analysis crossover point), JAX/XLA (staging artifact model, `static_argnums`), Diffrax (adaptive stepping vs tracing tension), Enzyme (reverse-mode performance ceiling), DiGeo (`HasCustomAdjoint` pattern for non-smooth discrete operations; parallel transport as geodesic byproduct), TaylorIntegration.jl (jet coefficient propagation), Mathlib4 (Lean proof foundation).

The trait hierarchy includes a `HasCustomAdjoint` trait covering DiGeo-style proxy gradients for discrete geometric operations, implicit function theorem adjoints for nonlinear

solves, continuous adjoint methods for ODE trajectory sensitivity, and any future operation where automatic AD is insufficient.



§ Status & provenance

Twelfth iteration; the v11 hybrid PRD ran 6,900 lines as "geological strata" that mixed graph-first and comptime-first framings. v12 is the rewrite around the hybrid architecture as primary. Produced in conversation [73f1973e](#) on 2026-03-20. Companion to the JAX and Mojo roadmap monographs (Nos. 03 and 04) which assume this architectural foundation.